

Challenges in Organizational Control: The Economic and Management Perspectives

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ABSTRACT

Organizational control is an important while challenging topic. Both the organizational economics and management literature have studied extensively the mechanisms of organizational control and have each arrived at important insights. However, these two literatures have developed largely separately and do not communicate with each other. By combining insights from both organizational economics and management literature, the authors assess the strengths and weaknesses of each of the three types of organizational control methods: incentives, monitoring, and social control. They conclude that as jobs nowadays are increasingly more complex, organizations need to combine the three control mechanisms to motivate their employees to achieve the cooperative behaviors and to fulfill the goals of the organizations.

Keywords: Incentives, Monitoring, Organizational Control, Organizational Economics, Social Control

INTRODUCTION

Firms face many challenges in motivating their employees and coordinating their activities. As Gibbons (2000) points out, “if Coase is correct that firms will exist only where they perform better than markets could, then it is a short step to argue that inefficiency inside organizations will be typical, not exceptional”. In other words, being better than markets does not mean that firms are always efficient. Although firms do better than markets in certain situations (Coase, 1937), there are many problems and difficulties in firms as well.

As Gibbons and Roberts (2013) argue, in circumstances where contracting is not a problem, markets work well and perhaps are better than firms. But familiar sources of contracting difficulties, such as pre-contractual private information leading to adverse selection, post-contractual asymmetric information or hidden action leading to moral hazard, or contractual incompleteness for whatever reasons, mean that market solutions may fail. But the managed organizations, with a broad array of levers to affect motivations and behaviors, are less disadvantaged than markets when contracting is beset by great difficulties or high transaction costs.

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The use of relational contract is one way that firms surpass the difficulties to write a complete and enforceable contract. According to the *Encyclopedia of Law and Economics* (Hviid, 2000), two features largely define what lawyers mean by a relational contract: incompleteness and longevity. Relational contracts govern continuing relations (Schwartz, 1992) and are in most cases long-term contracts (Goetz and Scott, 1980). The nature of the employment contract is often massively incomplete fraught with the complexities of time, uncertainty, and ambiguity. The employment relationship is a standard version of relational contract: First, it endures for some length of time; second, it is too costly or impossible to specify all contingencies in advance; third, the relationship and how it evolves facing new contingencies, is often characterized by its terms of governance, that is, which parties make which decisions.

Being an incomplete and relational contract gives employers much power in directing employees to do what employers want them to do. As Simon (1951) poses the question: Why is the employee willing to sign a blank check, by giving employer authority over his behavior? Also, the same questions are posed by Alchian and Demsetz (1972): “What ensures that the employee obey the employer’s instructions? What disciplinary power does a boss have that an independent contractor does not?” The employer, in many cases, acts to coordinate the activities of a number of employees and has the authority over what the employee does and the employee is directed by the employer about what to do. Under situations of incomplete contract, firms by using relational contracts and instructions can achieve better cooperation and prevent potential opportunistic behavior than market through using discrete contracts and prices.

Relational contracts are not the privilege of firms. Markets can also use relational contracts. But the relational contracts within firms are different from those between firms, as pointed out by Baker, Gibbons and Murphy (2001). Gibbons (2000) observes, “Firms are riddled with relational contracts – informal agreements that powerfully affect the behaviors of individuals

within the firm. Virtually every collegial and hierarchical relationship in organizations involves important relational contracts, including informal *quid pro quos* between co-workers and unwritten understandings between bosses and subordinates about task assignment, promotion, and termination decisions”. Explicit contract is enforced by law, but relational contract is usually enforced by trust and by reputation (Seppänen, 2014).

All organizations, as Arrow (1974) put it, share “the need for collective actions and the allocation of resources through nonmarket methods”. Managed organizations face all kinds of challenges in coordinating the behaviors and efforts of members and groups and in motivating them to perform the needed activities to fulfill the goals of the organizations. During recent years much attention has been devoted to organizational control problems within firms. Organizational controls are defined as the mechanisms utilized by managers to direct the behaviors and motivation of organizational members to act in desired ways to meet the organization’s objectives (Cardinal, 2001; Eisenhardt, 1985; Govindarajan and Gupta, 1985; Jaeger and Baliga, 1985; Kerr, 1985; Langfield-Smith, 1997; Ouchi, 1977, 1979; Snell, 1992). Critical to organization control is the problem of information (Holmstrom, 1982). Various articles have studied the design of organizational control mechanisms in order to overcome informational asymmetries in organizations. One must guard against an exclusive focus on fiat control, in which power is used to manipulate agents into action via a supreme authority of the principal (e.g., to fire the agent). There are several forms of control identified in the literature and it may be more instructive to think of control theory in terms of the capabilities to set up structures and rewards that motivate and influence rather than the colloquial notion of “control”.

The process of control is basically a process of monitoring, evaluating, and rewarding (Ouchi, 1977). From the perspective of organization design, the solution to this control problem is to shape the organization to bring a closer

alignment of interest between the organization and its members. In particular, to motivate employees, it is not just a matter of monetary rewards. They can also be other resources, as well as personal satisfaction or social approval and status. Different types of controls have been labeled: structural, also called bureaucratic or behavior control (Blau and Scott, 1962; Lebas and Weigenstein, 1986), input control (Merchant, 1985; Mintzberg, 1979, 1983), output control (Jaworski, 1988; Merchant, 1985), market control (Ouchi, 1979; Williamson, 1975), cultural control (Wanous, 1980), and integrative control (Roth, Sitkin, and House, 1994). In order to organize the discussion, we adopt Muralidharan and Hamilton's framework. Muralidharan and Hamilton (1999) identified three types of control systems in firms: output-based, behavior-based and clan/input-based. "The three types of control systems differ in how they seek to align organizational actions to stated objectives. Output-based systems focus on results, behavior-based systems focus on the process of work, and input-based systems focus on the antecedent conditions to performance, such as knowledge, skills, abilities, values, and motives of employees".

A similar typology is given by O'Donnell (2000), who identified three types of control methods: incentives, monitoring, and social control. Incentives are financial compensation based on some kind of performance; monitoring is activities or mechanisms used to obtain information about the employee behaviors; and social control is to create a culture or environment where employees can identify with the firm and therefore can internalize the values and goals of the firms.

Table 1 from the Appendix is adopted from Ouchi (1977). It shows under what kind of circumstances is each type of control mechanism applicable. In order to apply behavior-based control, the organization must possess some knowledge and agreement about the process through which efforts are transformed into outputs before managers can achieve control by monitoring and guiding the behavior of employees, as in the left two cells. On the other

hand, in order to apply output-based control, a reliable and valid measure of the desired outputs or performance must be available before managers can apply incentives based on the desired output, as in the top two cells. In the case when there is neither knowledge of the transformation process nor availability of output measures, organizations will have to rely on social control, in which they focus on the inputs to the production process, such as using personnel selection, training, rituals, creating a culture as means of control.

Of course, this formulation is only a normative one. It shows what types of controls are better suited for organizations under different circumstances, even though organizations may not always be rational. Indeed, in many situations, an organization may use one or multiple forms of control. In the following we will elaborate each of these three types of control and outline the challenges under each of them.

Both the economics and management literature have devoted significant attention to the study of organizational control. Since the organizational economics literature relies heavily on mathematical models, it has focused for the most part on incentives and monitoring, phenomena that can be more easily modeled by mathematical methods. But design robust and well-functioning tangible incentive mechanism can be very difficult when (1) workers engage in multiple tasks; (2) tasks are dynamic, interdependent among employees, and ambiguous; and (3) complex and new tasks need critical thinking, creativity and innovation. The management literature, on the other hand, relies more on empirical methods to observe what is happening in organizations and test the theories. Other than incentives and monitoring, it has emphasized the cultural, sociological and psychological aspects of organizational life, which are hard to model with mathematics. They include (1) an appeal to intrinsic motivation of employees to do a good job; and (2) an appeal to the employee internalizing the welfare of the employer or the interest of the organization (Baron and Kreps, 2013). Most of the time, these two streams of literature don't seem to be talking to each

other, even though they are studying the same phenomena. Therefore, it is the purpose of this paper to integrate both streams of literature, encompass them under a common framework and show how they complement each other.

OUTPUT-BASED/INCENTIVES

In fact, the output-based control in firms has its similarity to market mechanisms. Incentives can be designed to make individual compensation based on individual performance if it is measurable. In fact, pay for performance was fashionable and was one of the mantras of modern management for quite some time. Agency theory is commonly used to model this type of situations.

Encouraging employee efforts is a central issue in agency theory. When efforts are observable and contractible, providing efficient incentives is simple. However, in many situations efforts are either unobservable or non-contractible, which makes it impossible to simply pay the workers for their efforts. Most theoretical principal-agent model assume that the output is an increasing function of efforts and that it is also affect by some random variable. The principal can only observe the output, on which the compensation contracts are based. The problem for the principal (firm) is to set incentives compensation scheme to maximize its profit subject to the incentive compatibility and the participation constraints. The model has two implications. First, the stronger the incentives are, the harder the employee will work. Second, there is a trade-off between risks and incentives.

However, there are many problems resulting from pay-for-performance schemes which make the use of incentives less effective. Pay-for-performance schemes are problematic in multi-task jobs, where rewarding tasks whose performance is easier to measure will make the employees neglect other tasks whose performance are not easily measured or rewarded (Holmstrom and Milgrom, 1991). Pay-for-performance often makes employees

emphasize individual achievement rather than group or company goals. In other instances, upper level managers may not have the relevant knowledge or expertise to determine the appropriate outcomes. Pay for performance can also discourage innovation, because most rewards are based on current business, not exploration whose performance is hard to measure.

Kerr's (1975) article "On the Folly of Rewarding A, While Hoping for B" gives numerous examples of reward systems which reward performance that do not truly or complete reflect the whole spectrum of behavior or efforts that management wants to encourage. For example, in many universities, good teaching is hard to measure and reward, usually limited to the outstanding teacher awards, which are given to only a very small percentage of good teachers. Rewards for research and publications, on the other hand, are much more common and easier to measure and implement. Therefore, university professors often choose to concentrate more on research rather than on the teaching, especially in those research focused universities.

A similar example is, one of the key issues to proposed reforms of using incentive pay for teachers based on their students test score is that these incentives would have the effect that teachers would sacrifice higher level learning activities such as promoting curiosity and creative thinking and students' oral and written communication skills, and just focus on teaching the narrowly defined basic skills that are tested on standardized exams.

Furthermore, if the incentive is based on past performance, it can also lead to the so called "ratchet effect". Ratchet effect refers to the tendency of performance standards in an incentive system to be adjusted upwards after a particularly good performance, thereby penalizing good current performance by making it harder to earn future incentive bonuses (Milgrom and Roberts, 1992; Zhang, 1996). In most job situations, there is asymmetric information between the manager and the employees and the manager has few ways to know the abilities and cost of efforts of employees, and often has to set expectations based on employees' past per-

formance. In this kind of situation, employees may try to keep their efforts within a limit, so as to prevent the manager from deducing that the cost of efforts is low and further ratcheting up future performance expectations. The employees will have to balance between the desire to achieve good performance now and the danger of not being able to achieve good performance in the future under the increasingly higher expectations. Examples of these effects are sales people who are subject to sales quotas.

Team production is the norm in most organizations today, and in team production, individual performance is hard to measure, only team performance. Incentives are less effective when they are based on outcomes that result from interdependently performed tasks in teams, since it is difficult to link the incentives to the outcomes achieved by a specific individual (O'Donnell, 2000). One of the solutions to incentive problems in team production is a share contract. Shares contracts have been used to model contractual arrangements within and outside firms such as sales force compensation, franchising, licensing, legal fees, and author-publisher contracts. However, in most of the team situations in organizations, the division of labor is not clear enough and the collaboration is to a much closer degree to allow share contracts. In these situations, incentives based on individual performance become impossible and moral hazard is likely as individuals are liable to free ride. Holmstrom (1982) models the moral hazard in teams and suggests using competition and relative performance evaluation as a device to extract information optimally and prevent free riding.

Relative performance evaluation can be implemented in two different ways. On the one hand, firms can use the performance of a peer group of competitors to filter out shocks that are common to the whole peer group. This can lower the risk that the firms imposed on individual employees. On the other hand, firms can use fixed rewards for a fixed group of employees, where the distribution of rewards is based on the order of the participants' performance. This reward system is also known as a tournament,

can vary from a group of people compete for a single position or promotion to a set of prizes that diminish in value as a person's relative performance diminishes.

However, relative performance evaluation is not always available and often fuzzy. On the one hand, it is hard to compare the performance of different people who are assigned different tasks and type of jobs. On the other hand, there is no objective performance measurement for individuals in a team production setting. Thus, relative performance evaluation is in essence behavior-based monitoring method. To sum up, because it is often difficult or impossible to measure individual's performance and assess a worker's total contribution to firm value, pay-for-performance schemes are inevitably imperfect, which is why we often need the behavior-based monitoring method.

BEHAVIOR-BASED/ MONITORING

An alternative to financial incentives is to simply monitor workers. Indeed, monitoring and incentives are often viewed as substitutive or complementary mechanisms, with the balance between them determined by the trade-off in their respective costs or difficulties (Eisenhart, 1989). Alchian and Demsetz (1972) long ago pointed out that monitoring as a solution to incentive problems in team productions, as "transactions involving joint or team production require careful monitoring so that each actor's contribution can be assessed". Agency theory suggests that principal collects information about agent behavior, and thus limit the risk the agent will engage in behavior that is not in the interest of the principal (Holmstrom, 1979). One way to monitor employees is through bureaucratic mechanisms such as rules, programs and procedures (Galbraith, 1973), which set some standards to employees' actions and decisions.

However, behavior-based monitoring also has certain conditions for it to be effective. As mentioned before, the organization must possess some knowledge and agreement about the

process through which efforts are transformed into outputs before managers can achieve control by monitoring and guiding the behavior of employees. Thus behavior control is easier in simpler types of jobs, as for example in Taylor's time-motion study of assembly workers, but more difficult in more complex jobs that require more information processing, discretion and decision making (Fama and Jensen, 1983). Due to information asymmetry, often an employee is in a better position to evaluate to address the problem at hand, and it is hard for a supervisor to fully understand all the information associated with these problems. Thus, as the decision making discretion and information asymmetry increases, it becomes more difficult and less effective to use behavior based controls and monitoring.

Shapiro and Stiglitz (1984) propose that a firm pays employees efficiency wage or a wage premium if it cannot perfectly monitor workers. Employees, facing choice between earning this wage premium while working hard and facing a probabilistic chance of being caught and fired if they shirk, choose to work hard. Efficiency wage theory implies a negative relationship between monitoring intensity and wages. The empirical evidence for this implication is mixed. This could be many firms that can monitor very effectively or use output based incentives, while others use efficiency wage (Lazear and Oyer, 2013).

At high levels of interdependently performed tasks which need a lot of information sharing, coordination, creativity and innovation, and decision making, neither incentives nor monitoring is effective for managing employees. In fact, it has been suggested that monitoring, while it increases the cost and reduce the likelihood of employees engaging in opportunistic behavior, can on the other hand create negative feelings toward the organization, which may actually increase the opportunistic behavior (Ghoshal and Moran, 1996). The cooperative behaviors needed in complex jobs with closely collaborating teams are best facilitated through social control methods, so that individuals identify with the organization and internalize

and share the values, goals and interests of the organization as a whole (Ouchi, 1979). Research has shown that identification and internalization generate pro-social behaviors that promote the welfare of the organization rather than the individual.

INPUT-BASED/ SOCIAL CONTROL

Due to the often lack of information and thus difficulty to use incentives or monitoring, social control is preferred in many situations. In situations of asymmetric information, effective monitoring may be too costly. Furthermore, monitoring is less likely to be effective in eliciting organization citizenship behavior in which an employee thinks in terms of the goal of the organization. Meanwhile, when workers are intrinsically motivated to do a good job, emphasizing on financial incentives can dull or even reverse intrinsic motivations (Deci, 1975). Therefore, one way to reduce both the costs of incentives and those of monitoring is to rely on the intrinsic value workers get from doing their jobs well. If firms pick up workers carefully and foster a culture of commitment to reassure employees that their long-term interests are aligned with those of the companies they serve, and thus make employees internalize company objectives into their own objectives, the employees will be productive.

Input-based/social controls stress the antecedent conditions to performance, such as knowledge, skills, abilities, values, and motives of employees. Its aim is to foster a culture where employees identify with the firm's objectives and voluntarily engage in activities that are good for the firm.

The management literature has long recognized that worker motivation is not just derived from incentives and monitoring, but can be intrinsic as well. Early theories of work motivation can be divided into two opposing views at that time: one was the carrot and stick principle, which was a basic tenet of so-called Taylor scientific management and falls into the

monitoring category; in contrast, the other was the notion that a happy worker is a good worker. There are so many theories of what induces work motivation, which include motive/need theory, incentive/reward theory, reinforcement theory, goal theory, personal and material resource theory, group and norm theory, socio-technical system theory, expectancy-value theory, equity theory, attitude theory, intention/goal theory, attribution/self-efficacy theory, and other cognitive theories such as the social information processing theory (Katzell and Thompson, 1990; Zhang, Wang, Colucci, and Wang, 2014).

Take the example of intrinsic motivation, which certainly has an important effect on work efforts of employees. Intrinsic motivation is motivation derived from the perception of rewards inherent in the task itself, as opposed to extrinsic motivation, which is motivation derived from extrinsic rewards, such as monetary rewards. Many activities such as manipulation, exploration, and information processing provide satisfaction in and of themselves. Hackman and Oldham (1976)'s job characteristics model is one influential theory of work motivation, which focuses on the intrinsic aspects of the job. They identify five job characteristics that contribute to intrinsic work motivation, which are skill variety, task identity, task significance, autonomy, and feedback. The important implications are that improving job design can increase work motivation and then work efforts.

According to the expectancy-value theory in organization behavior literature, intrinsic motivation and extrinsic motivation are generally considered additive in making up total motivation. However, there are also exceptions as in some cases, extrinsic rewards may actually hurt or reduce intrinsic motivation (Kreps, 1997). For example, Deci (1971) found in his experiment on college students that giving students money actually decreased their intrinsic motivation in playing a very interesting game. Staw and Calder (1980) suggests that when there is no norm for payment, e.g. in many voluntary task activities such as political, charity, and welfare work, introducing pay into an interesting task may decrease intrinsic motiva-

tion. Yet in industrial organizations, most jobs are not inherently interesting enough to foster high intrinsic motivation and people usually hold beliefs that they ought to be paid, and in these situations, it is more likely that extrinsic rewards will have a positive effect on intrinsic motivation (Staw, 1989).

Organizational economists have long tried hard to figure out the most efficient way to motivate employees in organizations. However, most of the economic models have been focusing on incentive design and rely on stringent conditions of information availability, which are often simply too difficult or costly to collect in organizations. Few economic models have incorporate intrinsic motivation as a factor on work efforts, although intrinsic motivation has an important impact on work motivation. To enrich our understanding of organizational control, greater attention needs to be paid to the sociological and psychological aspect of work motivation (Baron, 1988). Things like norms and customs, social comparison, peer pressure, job prestige, human relations, etc. all have significance in motivating employees.

Fortunately, however, recently there have been some efforts to incorporate these other psychological or sociological factors into economic models of work efforts. Lazear (1992) adds peer pressure to the model on incentives; Rotemberg (1994) studies the implications of human relations to work efforts; Akerlof and Kranton (2000) incorporates identity into economic decisions. Murdock (2002) studies the role of intrinsic motivation on optimal incentive contracts. Hart and Holmstrom (2010) pay attention to human assets in their broadened approach to the theory of the firm, taking into account the fact that workers care about money and the type of job they do (they get private benefits from actions) and look at the various implications of this to the firm structure, e.g. optimal matching of workers with types of bosses/firms.

CONCLUSION

Both the organizational economics and management literature have studied the mechanisms of organizational control extensively, and in doing so they have each arrived at important insights. The economics literature is more concerned with the general theory of how incentive and monitoring structures can be used in the coordination of economic activities, while the management literature is more concerned with the empirical reality of how different organizations are managed in practice. However, these two literatures have developed largely separately, and so they have not talked much to one other.

By combining insights that are originally two separate streams of literature, this article is able to incorporate the respective strengths of each of these strands of literature, and to emphasize how their perspectives may complement one another. The economics literature has made substantial progress in the formal modeling, which has given rigor to the study of organizational control, while neglecting some of the more fuzzy or ambiguous aspects of organizational control. In contrast, the management literature has been grounded on empirical observation and testing, and thus is better able to incorporate detailed evidence on what is happening in the field, but at the same time might be usefully complemented by the rigorous modeling techniques more familiar in organizational economics.

After assessing the strengths and weaknesses of each of the three types of organizational control methods, we come to realize that increasingly organizations need to rely on social control methods. Organizational control is an extremely important and at the same time challenging topic. Incentives and motivation, when done the right way, are powerful in generating the most energy and unleashing the creativity out of people. Yet getting them right is often very difficult due to the many problems involved, such as difficulty of performance measurement, information asymmetry, team production, and complexity of jobs.

The smarter organizations today need to better utilize social control methods to get the most productivity and creativity out of their employees. We have seen some excellent examples in this case, such as Google creating a great relaxed working environment to its employees, providing snacks, massage, child care, and giving its engineers up to 20 percent of their work time to experiment and work on their own ideas and innovate.

Jobs nowadays are increasingly more complex and multifaceted, and thus relying on monitoring and incentives as control methods are becoming increasingly insufficient. In many situations social control is often the best complement to achieve the needed cooperative behaviors in a complex job situation.

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APPENDIX

Table 1. Organizational control mechanisms

	Knowledge of Transformation Processes		
		Good	Poor
Availability of Output Measures	High	Behavior-Based Control or Output-Based Control (e.g. tin can plant)	Output-Based Control (e.g. Life insurance agency)
	Low	Behavior-Based Control (e.g. double play combination)	Social Control (e.g. Foreign service)